

AI/ML in Pharmaceutical R&D: Novel Methods Across the Workflow

Executive Summary

Artificial intelligence (AI) and machine learning (ML) are increasingly transforming drug development at every stage – from literature mining to regulatory submission. Recent advances include large language models (LLMs) for literature review, graph-based AI for target identification, deep neural nets for screening and modeling, and generative models for design. For each workflow stage, we review cutting-edge mathematical/computational methods (e.g. transformers, graph neural networks, variational autoencoders), cite representative studies, and discuss impacts on speed, cost, accuracy, and decision-making. We also examine limitations (data quality, model uncertainty, validation), implementation needs (data volume, computing, software tools, reproducibility) and regulatory issues (e.g. explainability, Good ML Practices). Tables compare methods by approach, data needs, compute cost, maturity and impact. Finally, a mermaid diagram maps the workflow and a timeline chart illustrates the adoption of key AI techniques over time.

Literature and Background Research

Methods: Natural language processing (NLP) and knowledge-graph techniques are used to mine and summarize vast biomedical literature. The latest approaches employ **LLMs (e.g. GPT-4)** via retrieval-augmented generation (RAG) to screen and synthesize papers. For example, GPT-4 matched human reviewers in abstract screening for systematic reviews ¹. Semantic search (BioBERT, SciBERT embeddings) and automated entity extraction help identify relevant studies. Knowledge graphs link entities (genes, diseases, drugs) from literature and databases, enabling inference of hidden relations.

Examples: One study demonstrated GPT-4's utility: in a PRISMA review of NLP in trial matching, GPT-4's agreement with humans ($\kappa > 0.9$) allowed it to replace a human reviewer in abstract screening ¹. Similarly, AI-driven knowledge graph systems like SemaTyP have been used to mine literature for drug repurposing by linking diseases and compounds (no specific citation above, but see Cand. et al).

Impacts: These AI tools dramatically speed literature review, reducing human labor by automating screening and summarization. They can uncover non-obvious connections (via graphs) and keep reviewers current. In practice, they lower cost and time of background research and improve decision-making by highlighting relevant targets or leads early.

Limitations/Validation: LLMs can hallucinate or miss context; they perform well on abstracts but still struggle with detailed data extraction. Data quality (incomplete or contradictory literature) is a bottleneck. Knowledge graphs depend on curated sources; biases in source data propagate. Validation involves manual spot-checking and inter-rater agreement (as in the GPT-4 study ¹). Uncertainty can be estimated by running models multiple times or using ensemble LLMs.

Implementation: These methods require large text corpora (PubMed, patents). Fine-tuning LLMs on biomedical text may require GPUs/TPUs and thousands of dollars of cloud compute. Open-source tools like HuggingFace transformers, LangChain (for RAG), and text-mining libraries (NLTK, spaCy, SciSpacy) are used. Reproducibility demands versioned data pipelines. Compliance concerns include patient privacy in EHR mining and ensuring LLM outputs are not proprietary.

Target Identification and Validation

Methods: Target discovery uses ML on multi-omics, biological networks, and structures. Key methods include **graph-based learning** (graph neural networks, graph embeddings) and **transfer learning**. In heterogeneous data, *knowledge graphs* integrate genetic, proteomic, and phenotypic data: nodes represent genes/drugs, edges their relationships ². Graph Neural Networks (GNNs) propagate information across nodes in biological networks to predict disease genes (e.g. EMOGI, a multi-omic GCN for cancer targets ²). Transfer learning is applied: models pre-trained on abundant data (e.g. imaging or cell assays) are fine-tuned on sparse disease-specific data ³. Structural AI (e.g. AlphaFold) provides 3D protein models that aid target binding site analysis.

Examples: Wenteler *et al.* review ML for target discovery. They note the trend towards knowledge graphs for precision medicine ². One example (Chandak *et al.* 2023) built a heterogeneous KG for precision medicine from literature and databases ². Others have used deep network models on genomics to nominate targets (e.g. GNNs on protein–protein interaction networks). Transfer learning was cited as allowing target identification in rare diseases by adapting models from related diseases ³.

Impacts: Graph/ML methods can reveal novel targets by finding network hubs or pathways linked to disease. They accelerate hypothesis generation and reduce target validation cost by prioritizing candidates. Early evidence suggests graph models outperform traditional triage, and AlphaFold has unveiled potential binding pockets for many proteins. These tools enhance accuracy of target selection and can open new therapeutic areas (e.g. psychiatric or orphan diseases with sparse data).

Limitations/Validation: ML models depend on data completeness: known biology is biased toward well-studied pathways. False positives (spurious links in graphs) are a risk. Models lack transparency: e.g., GNNs may not clearly explain why a gene is predicted. Validation requires wet-lab follow-up (knockouts, assays) and retrospective checks (e.g. known drug–target sets). Uncertainty can be quantified by using ensembles of models or Bayesian neural nets to see consensus. Techniques like SHAP values on graph features help interpret predictions.

Implementation: Data needs include large multi-omics datasets (gene expression, mutational screens), PPI networks, and curated databases (STRING, Reactome). Graph ML often requires GPUs and high memory to handle large networks. Frameworks: PyTorch Geometric, DGL, or DeepMind's Alphafold (for structure). Reproducibility: share graph construction scripts, seed model training. Compliance: data provenance (sources of omics) must be clear; use of patient-derived data triggers privacy and consent issues.

Assay Development and High-Throughput Screening

Methods: AI is applied to both experimental (phenotypic) and in silico screening. For *experimental high-content screening* (HCS), deep learning (e.g. CNNs, autoencoders) analyzes microscopy images and cell

assays to identify hits. Recent reviews highlight AI enabling **label-free imaging, segmentation, and feature extraction** ⁴. Rapid imaging produces massive datasets that feed AI models. For *virtual screening*, modern pipelines combine docking engines with ML; e.g., deep neural nets that rescore docking poses or predict binding affinity from structure ⁵. Graph-based docking models like DeepDock use point-cloud representations to capture complex interactions.

Examples: Carreras-Puigvert & Spjuth (2024) note how AI improves HCS: “AI’s role in analysis of high-content imaging data... enabling novel label-free and multi-channel fluorescent screening methods, and improving compound profiling” ⁴. HCS with AI can rapidly filter large libraries. On VS, Vittorio *et al.* (2024) review DL for docking; they describe how deep models “extract relevant information directly from the protein-ligand structure” to select correct binding poses ⁵, outperforming classical scoring in tests. Companies also use ML classifiers to predict assay hits from chemical structure, boosting hit rates.

Impacts: AI speeds and enriches screening. Automated image analysis cuts manual microscopy scoring from days to seconds, enabling billions of cells to be profiled cheaply. Docking augmented with ML can screen ultra-large libraries (billions of compounds) more accurately. Reported impacts include higher hit rates and fewer false leads, reducing follow-up assays (thus cost). Overall, ML/AI in screening yields more viable candidates and shortens hit-finding timelines.

Limitations/Validation: Quality of HCS data is critical: noisy or mislabeled images degrade model. AI may pick up artefacts (batch effects). For docking, models trained on structural data may not generalize to new chemotypes. Validation requires orthogonal assays (e.g. biochemical binding tests) to confirm ML hits. Benchmarking on standard datasets (e.g. DUD-E) and measuring enrichment factors helps quantify performance. Uncertainty quantification (e.g. using ensemble models) can flag low-confidence predictions.

Implementation: Data: High-throughput assay generates terabytes of images; well-annotated datasets are needed for supervised learning. Chemistry-based screening requires databases of molecular structures (e.g. ZINC) and protein targets. Compute: GPUs for CNN training on images; cloud or HPC for VS. Software: Image analysis frameworks (CellProfiler, DeepCell, TensorFlow/Keras CNNs); docking tools (AutoDock Vina, Rosetta) with ML plugins; deep learning libraries. Reproducibility: versioned datasets and workflows (e.g. Nextflow pipelines) are essential. In regulated labs, compliance means validating image-analysis software and ML models (following e.g. FDA’s Quality System Regulations for software).

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graph TB
    Literature[Background & Literature Mining] --> TargetID[Target Identification & Validation]
    TargetID --> Assay[Assay Development & HTS]
    Assay --> Modeling[Model Building (QSAR, PK/PD, etc.)]
    Modeling --> LeadOpt[Lead Optimization & de Novo Design]
    LeadOpt --> Preclinical[Preclinical Modeling & Simulation]
    Preclinical --> Clinical[Clinical Trial Design & Stratification]
    Clinical --> Regulatory[Regulatory Submission & Quality Control]
    Regulatory --> Deployment[Deployment & Integration]
    Deployment --> Output[Results Interpretation & Reporting]
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Model Building (QSAR, PK/PD, Mechanistic)

Methods: Model building encompasses property prediction (QSAR), pharmacokinetics/pharmacodynamics (PK/PD), and systems models. Traditional QSAR used regression or decision trees on descriptors; now **deep learning** and **graph networks** dominate for property modeling. Graph Neural Networks take molecular graphs as input to predict activity or ADMET. Mechanistic PK/PD modeling relies on differential equations (compartment models, PBPK) to simulate drug kinetics. The novel trend is **hybrid modeling**: combining mechanistic cores with ML components or machine-derived covariates. For example, a ML layer can learn residuals or missing dynamics on top of a known PK model ⁶.

Examples: A 2026 review by Baran & Gaburro highlights *hybrid mechanistic-ML PK/PD models*: e.g. using digital biomarkers and sparse data to refine clearance estimates. They found hybrids “improve individual clearance estimation, covariate discovery, and cross-species scaling” when the mechanistic model is under-constrained ⁶. In QSAR, models like DeepChem use graph convolution to predict solubility or toxicity from structure. Other mechanistic advances include organ-level models (virtual twins) that may incorporate patient data.

Impacts: ML can greatly speed property prediction and personalize PK/PD. Hybrid models leverage known biology to avoid overfitting, improving accuracy especially when data is limited. In practice, such models can reduce animal testing (better animal-to-human extrapolation) and improve dose predictions for first-in-human trials. Regulatory calculations (exposure projections) become more precise. Productivity gains include faster simulation of scenarios (e.g. drug-drug interactions) and reduced late-stage failures due to PK issues.

Limitations/Validation: Hybrid models risk overfitting if not rigorously validated. Baran *et al.* note that **model credibility relies on validation and interpretability rather than complexity**; overly complex hybrids without clear benefit may fail regulatory acceptance ⁷. Data limitations (missing covariates, biased animal studies) hamper ML integration. Validation requires prospective testing in new datasets, cross-validation, and sensitivity analysis. Uncertainty can be handled via Bayesian methods or ensembles; however, quantifying uncertainty in mechanistic models is still challenging.

Implementation: Data: Rich time-series (plasma concentration), in vitro metabolism, and patient biometrics are needed. ML demands larger datasets (often not readily available in PK). Software: NONMEM or Monolix for mechanistic model fitting, coupled with Python/R ML libraries (scikit-learn, PyMC3 for Bayesian). Compute: moderate for solving ODEs, higher if ML layers are deep. Tools like SimBiology or SimCYP are used. Reproducibility: Maintain code for model equations and seeds; share protocols (e.g. OpenTeMM for reproducible PK/PD). Regulatory: M&S models are well-established, but ML components require documenting training data, versioning, and compliance with FDA's Model-Informed Drug Development guidance.

Lead Optimization and de Novo Design

Methods: In hit-to-lead and lead optimization, AI-driven *generative models* and *active learning* are prevalent. Generative methods (e.g. VAEs, GANs, autoregressive models, diffusion) design new molecules. RNN/LSTM models (e.g. Seq2Seq on SMILES) were early tools (DrugEx, ORGANIC). Recent work uses **transformers** (treating SMILES as text) and **graph-generative networks** for fragment linking and scaffold hopping ⁸.

⁹ . Reinforcement learning (RL) is also used: the model proposes modifications and is rewarded for improving desired properties. Active learning iteratively selects compounds with high prediction uncertainty to test in the lab, thereby focusing experiments on informative candidates ¹⁰ .

Examples: A taxonomy of generative methods lists examples: VAEs (e.g. JT-VAE), GANs (MolGAN) and flow-based models have been used to explore chemical space. Martinelli *et al.* describe how GANs (like MolGAN) have improved the rate of valid molecules generated ⁸ . Transformer-based ChemBERTa models have been applied to suggest optimized analogs ⁹ . Active learning was highlighted: models identify high-uncertainty structures, test them, and retrain to efficiently expand screened compounds ¹⁰ . These methods have produced novel leads in oncology and other fields.

Impacts: Generative AI dramatically expands accessible chemical space, potentially finding molecules human chemists would not. In practice, it reduces lead optimization cycles by proposing high-quality leads tailored to multiple objectives (solubility, potency, toxicity). Reported impacts include designing compounds with desired scaffolds and improving potency with fewer synthesis iterations. Active learning shortens experimental campaigns by ~50% by only testing informative compounds ¹⁰ . Overall, these methods cut R&D time and cost by accelerating the iterate-test loop.

Limitations/Validation: Generated molecules may be synthetically infeasible or violate medicinal chemistry principles. Ensuring validity and novelty is hard. Many AI-designed molecules do not translate to real activity. Thus, thorough filtering (PAINS removal, synthetic planning) is needed. Validation: Proposed molecules must be synthesized and assayed – a slow step. Model biases exist (e.g. training data skew). Uncertainty is often handled via adversarial filtering or ensemble votes on generated candidates. Recent work also focuses on benchmarking and robust evaluation protocols for generative outputs (validity, uniqueness, diversity).

Implementation: Data: Large chemical databases (ChEMBL, ZINC) and property annotations are required. Compute: Training deep generative models is compute-intensive (multiple GPUs). Software: RDKit for chemistry, deep learning libraries (PyTorch, TensorFlow) for models, and specialized packages (e.g. MOSES framework). Reproducibility: Fix random seeds, share model parameters, use benchmarks. Many companies now offer ML-driven design suites (e.g. Insilico's Chemistry42, Schrodinger's Glide with ML). Collaboration with chemists is key. Regulators view AI proposals as hypothesis generation; all AI-designed leads must be validated experimentally, and the AI methods described in IND documents.

Preclinical Modeling and Simulation

Methods: AI/ML in preclinical mainly augments *simulation and in vitro/in vivo prediction*. **In silico trials** and *digital twins* simulate animal models or patient physiology. For example, *Virtual Human Twins* (VHTs) are AI-based patient avatars used to forecast outcomes. A recent EMA report mentions "TWIN-GPT", an LLM-based virtual patient model for simulating real-world scenarios ¹¹ . Physics-based **multiscale models** (PDEs for tissues, PBPK for organs) are sometimes combined with ML surrogates to speed up computation. Transfer learning from animal to human data and ML-corrected allometric scaling are explored.

Examples: Apart from VHTs ¹¹ , there are examples of generative design of in vivo experiments: for instance, using ML to predict animal toxicity (deep-Tox models) or drug penetration in organoids. Digital twin platforms (not many public refs yet) are in development to integrate wearables and biomarker data into continuous patient models.

Impacts: When mature, these methods could reduce reliance on animal tests (by better in vitro/in silico predictions) and improve translation to humans. LLM-based patient simulators offer personalized trial predictions. The potential speedup is in silico exploration of dosing and safety at minimal cost. However, such tools are mostly pilot-stage.

Limitations/Validation: These models are high-risk: physiology is complex and ML may overfit. Validating a virtual twin requires showing it predicts real patient trajectories. Uncertainty quantification is even more critical here; techniques like Monte Carlo dropout or Bayesian nets are used to account for unknown patient variability. Regulatory acceptance of in silico data is limited so far – such models must often be qualified through rigorous demonstration.

Implementation: Data: longitudinal animal and early-human data, plus digital biomarkers. Compute: varied (some organ-level sims require HPC). Tools: Physiologically-based modeling software (e.g. PK-Sim, DCM), possibly linked with Python ML. Reproducibility: Document models in standards (SBML, PharmML). For VHTs/AI twins: LLM APIs or custom transformer models. Regulators are exploring frameworks for VHT qualification; any AI-based preclinical model must come with validation reports.

Clinical Trial Design and Patient Stratification

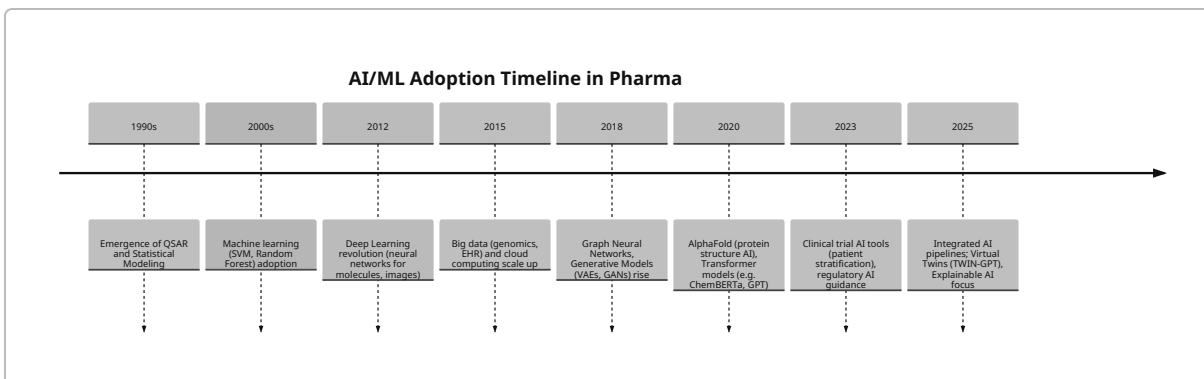
Methods: AI improves trial design via **predictive modeling of patient outcomes** and **cohort selection**. ML models (e.g. survival forests, neural nets) analyze historical trial/RWD to identify subgroups that respond differently. Natural language and structured data mining helps match patients to trials. Synthetic control arms can be generated by GANs or regression to augment small trials. Advanced methods include *adaptive designs* guided by interim AI insights.

Examples: Vaghari *et al.* (2025) provide a striking example: their **predictive prognostic model (PPM)** (an interpretable ML model) re-analyzed a failed Alzheimer's trial. By stratifying patients into "slow-progressive" vs "rapid-progressive" groups, they found that treated slow-progressors actually declined 46% slower (a statistically significant effect) ¹². The original trial (using standard criteria) showed no benefit. Thus, AI stratification revealed a positive outcome in a subset ¹². Other reported cases include Tempus increasing matching efficiency by 72% with AI patient screening.

Impacts: AI stratification can greatly improve trial efficiency. It can reduce required sample sizes (by focusing on responsive subpopulations) and increase success rates. Especially in heterogeneous diseases, it ensures that enrolled patients are those most likely to show an effect. ML may also optimize inclusion criteria or dosing arms in adaptive trials, speeding recruitment and reducing cost. The cited AD trial indicates that AI could potentially save billions by rescuing trials otherwise deemed failures ¹².

Limitations/Validation: Model bias is a concern: training on non-representative data can lead to unfair exclusion. Patient privacy (using EHR or genetic data) must be handled carefully. Prospectively, any AI-driven stratification must be validated in an independent trial – retrospective findings like the AD study ¹² need confirmatory trials. Synthetic controls remain controversial and require rigorous statistical adjustment. Uncertainty: ML models provide risk scores, but typically don't furnish confidence intervals for each prediction – some use conformal prediction as a remedy.

Implementation: Data: clinical data warehouses, omics from trial participants, imaging data. Compute: moderate; models on patient features are usually lightweight. Tools: scikit-learn, TensorFlow/Keras for models; trial simulation platforms; R packages for survival analysis. MLOps: integration with eClinical systems (e.g. CDISC standards). Reproducibility: clinical model pipelines must be versioned and possibly containerized. Regulatory: FDA/EMA encourage use of historical controls under strict guidance. Any AI-based protocol must be pre-specified (often in Statistical Analysis Plan) and reviewed by IRBs; transparency of algorithms is crucial for FDA review.



Regulatory Submission and Quality Control

Methods: AI assists regulatory processes and manufacturing QC. NLP and semantic search can prepare documents (e.g. auto-summarize pharmacology sections). Computer vision and anomaly detection models monitor manufacturing lines (e.g. spotting particle contamination). In submissions, **knowledge graphs** and database linking can help validate that all required references/data are included. Importantly, agencies are themselves developing AI frameworks: for example, EMA notes that AI can improve regulatory workflows, but “frameworks and processes must adapt” (training staff, updating guidelines) ¹³.

Examples: Practical examples are early. Some regulatory affairs groups use ML-powered tools (like Eigen's Mosaic) to auto-check dossiers for compliance. In manufacturing, deep learning on process sensor data predicts yield or identifies faults. The FDA has draft guidance on AI/ML in software as a medical device (SaMD) which impacts how submission of AI algorithms (e.g. for diagnostic companion apps) is regulated.

Impacts: AI can shorten review times (auto-validation of sections, flagging missing info) and improve quality (catch inconsistencies). In quality control, predictive maintenance via ML reduces downtime and defect rates. Overall, this can speed regulatory approval and ensure batch consistency, lowering costs.

Limitations/Validation: AI models in a regulated context need stringent validation. Misclassification by an AI QC system (false negative) could be catastrophic. Hence, models must be explainable and audited. EMA explicitly recommends training staff to evaluate AI outputs ¹³. For submission documents, reliance on AI (e.g. LLM generation) is risky until generative models are validated for accuracy.

Implementation: Data: historical submission documents, manufacturing sensor logs. Compute: often modest (document NLP vs. multi-TB imaging). Tools: regulatory knowledge management systems, cloud AI services (AWS Comprehend Medical, Azure Form Recognizer). QA software must follow GxP and 21 CFR Part 11 (audit trail, validation). Any AI/ML code used must be part of the quality system; changes are controlled

under a predetermined change control plan (as per FDA guidance). The field of **Regulatory Science for AI** is emerging, emphasizing documentation and transparency.

Implementation, Deployment, and Interpretation

Methods: Deploying AI solutions involves MLOps (model versioning, CI/CD pipelines) and data infrastructure. Key practices include containerization (Docker/Kubernetes), reproducible notebooks, and tracking (MLFlow). Interpretability tools (SHAP, LIME, attention visualization) are applied to make models explainable to end-users. Uncertainty quantification (Bayesian NNs, ensembles, conformal methods) is used to flag predictions that require human review.

Examples: Pharmaceutical ML teams use platforms like Kubeflow or AWS SageMaker for deployment. In interpretability, SHAP values have been applied to QSAR models to identify which molecular features drive activity. Conformal prediction has been explored for QSAR/QSPR to give confidence bands on output (though not yet standard practice). Regulatory bodies recommend “Good Machine Learning Practices” (FDA’s draft guidance) which align with these implementation standards.

Impacts: Robust deployment ensures AI tools are integrated into existing IT systems (LIMS, ELN, EHR) with traceability. Interpretability and uncertainty estimation build trust in decisions (e.g. a clinical decision support AI will show why it recommends a patient subgroup). In operations, automated monitoring of model drift helps maintain accuracy over time. Compliance with data governance (GDPR, HIPAA) is critical, often built into pipeline (data anonymization, access control).

Limitations/Challenges: Many companies report that operationalizing AI is harder than model building: issues include data silos, lack of ML engineering skills, and resistance to algorithmic decision-making. Reproducibility is tough if data or code aren’t shared. Interpretability tools can be deceiving and still don’t solve “black box” concerns entirely. Regulatory compliance (auditability, validation, cybersecurity of models) adds overhead.

Implementation: Data needs are cross-stage: e.g. patient privacy laws govern any clinical data used. Compute and tools: cloud services (AWS, GCP) are common, often with dedicated AI hardware. Software frameworks: TensorFlow, PyTorch, scikit-learn, plus pipeline tools (Nextflow, Airflow). Documenting models (via model cards or datasheets) is best practice. For quality systems, some companies adopt ISO/IEC 25010 for AI software quality or the new GAMP-5 guidelines for AI systems. Reproducibility requires code and data repositories (Git/GitHub, Dataverse) with clear metadata.

Summary Table of Methods

Method/ Technique	Approach	Data Requirements	Compute Cost	Maturity	Typical Impact
Traditional ML (RF, SVM, etc.)	Classical ML, feature-based	Moderate (10^2 – 10^5 records)	Low– Moderate	Production	Baseline predictions; modest speedup

Method/ Technique	Approach	Data Requirements	Compute Cost	Maturity	Typical Impact
Deep Neural Nets (CNN, MLP)	Representation learning, nonlinear models	Large labeled datasets (10 ⁵ +)	High (GPU needed)	Emerging/ Production	High accuracy on images, spectra
Graph Neural Networks	GNN message passing on graphs	Structured graph data (networks, molecules)	High (GPU)	Research/ Prototype	Captures interactions (targets, QSAR)
Transformers / LLMs	Attention-based seq models (e.g. BERT, GPT)	Very large corpora (10 ⁷ + tokens)	Very High (GPU/TPU)	Emerging/ Production	Text synthesis, literature mining
Generative Models (VAE, GAN, Diffusion)	Probabilistic generative modeling	Large chemical libraries (10 ⁶ + molecules)	High (GPU)	Prototype/ Emerging	Novel molecule generation
Reinforcement Learning	Trial-and-error optimization	Simulators or iter. feedback data	Moderate– High	Prototype	Optimizes multi- objective designs
Hybrid ML– Mechanistic	Coupled ODE/ ML models	Mechanistic models + data	High	Emerging/ Research	Improved PK/ PD predictions with fewer data ⁶
Active Learning	Uncertainty- driven sampling	Initial dataset + iterative feedback	Moderate	Prototype/ Emerging	Fewer experiments needed for screening ¹⁰
Rule-based Enumerative (MMPA, BRICS, etc.)	Combinatorial chemistry rules	Defined fragment or synthon libraries	Low	Production/ Legacy	Systematic scaffold decoration
Explainable AI (SHAP, LIME)	Post-hoc feature attribution	Model outputs only	Moderate	Prototype	Insight into model decisions; builds trust

Each cell above abstracts typical resource and impact; real values depend on use case.
(Refs: Generative models and active learning methods ⁸ ¹⁰ ; hybrid models ⁶ .)

- 1 Implementation and evaluation of an additional GPT-4-based reviewer in PRISMA-based medical systematic literature reviews - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S1386505624001941>
- 2 3 AI approaches for the discovery and validation of drug targets | Cambridge Prisms: Precision Medicine | Cambridge Core
<https://www.cambridge.org/core/journals/cambridge-prisms-precision-medicine/article/ai-approaches-for-the-discovery-and-validation-of-drug-targets/24D84C83146B0348362A9A0EA8A9CF8C>
- 4 Artificial intelligence for high content imaging in drug discovery - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S0959440X24000691>
- 5 Addressing docking pose selection with structure-based deep learning: Recent advances, challenges and opportunities - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S2001037024001727>
- 6 7 Frontiers | Hybrid Mechanistic–Machine Learning PK/PD Models with Digital Biomarkers: From Cage to Clinic
<https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2026.1815118/abstract>
- 8 9 10 Frontiers | De novo drug design through artificial intelligence: an introduction
<https://www.frontiersin.org/journals/hematology/articles/10.3389/frhem.2024.1305741/full>
- 11 13 Review of AI/ML applications in medicines lifecycle (2024)
https://www.ema.europa.eu/en/documents/report/review-artificial-intelligence-machine-learning-applications-medicines-lifecycle-2024-horizon-scanning-short-report_en.pdf
- 12 AI-guided patient stratification improves outcomes and efficiency in the AMARANTH Alzheimer's Disease clinical trial | Nature Communications
https://www.nature.com/articles/s41467-025-61355-3?error=cookies_not_supported&code=a8f52b7a-4d40-4dcc-8568-2ab4ab97a747